

$$\begin{aligned} \underline{U}_L - \underline{U}_C - \underline{U}_{R3} &= 0 \\ \underline{U}_L &= \underline{U}_C + \underline{U}_{R3} \\ I_1 + I_2 + I_3 &= 0 \\ I_1 &= -I_2 - I_3 \end{aligned}$$

$$E = j100$$

$$X_L = 10 \rightarrow Z_L = 10e^{j90}$$

$$R = 10$$

$$X_C = 10 \rightarrow Z_C = 10e^{-j90}$$

$$\underline{U}_{R3} = R_3 I_3$$

$$\underline{U}_C = Z_C \cdot I_3$$

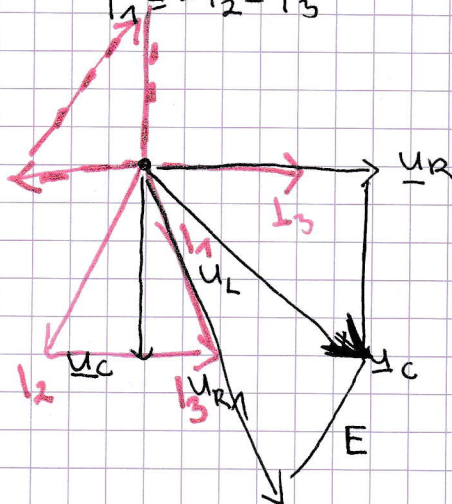
$$\underline{U}_L = Z_L \cdot I_2 \rightarrow I_2 = \frac{\underline{U}_L}{Z_L}$$

$$V_1 \left(\frac{1}{R_1} + \frac{1}{Z_L} + \frac{1}{R_3 + Z_C} \right) = \frac{E}{R_1}$$

$$V_1 \left(\frac{1}{10} + \frac{1}{10e^{j90}} + \frac{1}{10 + 10e^{-j90}} \right) = \frac{100j}{10}$$

$$V_1 \left(\frac{1}{10} + \frac{e^{-j90}}{10} + \frac{1}{10\sqrt{2}e^{-j45}} \right) = 10j \quad (\text{FOFO})$$

$$V_1 (1 + e^{-j90} + e^{j45}) = 10j$$



$$\underline{I}_1 = 4,47e^{-j116}$$

$$\underline{I}_2 = 6,32e^{j18}$$

$$\underline{I}_3 = 4,47e^{j153}$$

$$E = 100e^{j90}$$

$$\underline{U}_{R1} = 44,72e^{-j116}$$

$$\underline{U}_L = 63,24e^{j108}$$

$$\underline{U}_C = 44,7e^{-j63}$$

$$\underline{U}_{R3} = 44,7e^{j153}$$

$$4,47e^{-j116} + 6,32e^{j18} + 4,47e^{-j153}$$

